Heming Xia

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Education

The Hong Kong Polytechnic University **Jan. 2024 – Dec. 2026**
Ph.D. in Computer Science *Hong Kong, China*
Advisor: Prof. Wenjie Li
Thesis: Towards Lossless Inference Acceleration of Large Language Models

University of California, San Diego **Jan. 2026 – Apr. 2026**
Visiting Scholar in Computer Science *San Diego, United States*
Advisor: Prof. Julian McAuley

Peking University **Sep. 2020 – Jul. 2023**
Master in Software Engineering *Beijing, China*
Advisor: Prof. Zhifang Sui
Thesis: Speculative Decoding in Neural Machine Translation

Peking University **Sep. 2016 – Jul. 2020**
B.S. in Physics (Department of Astronomy) *Beijing, China*
Advisor: Asst. Prof. Lijing Shao
Thesis: Improved Deep Learning Techniques in Gravitational-wave Data Analysis

First-Author Publications

- **ToolSpec: Accelerating Tool Calling via Schema-Aware and Retrieval-Augmented Speculative Decoding**
  Heming Xia, Yongqi Li, Cunxiao Du, Mingbo Song, Wenjie Li
  Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing. ***EMNLP 2026****.*
- **Merlin's Whisper: Enabling Efficient Reasoning in Large Language Models via Black-box Persuasive Prompting**
  Heming Xia, Cunxiao Du, Rui Li, Chak Tou Leong, Yongqi Li, Wenjie Li
  The 64th Annual Meeting of the Association for Computational Linguistics. ***ACL 2026****.*
- **TokenSkip: Controllable Chain-of-Thought Compression in LLMs**
  Heming Xia, Chak Tou Leong, Wenjie Wang, Yongqi Li, Wenjie Li
  Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing. ***EMNLP 2025****.*
- **Beyond Single Frames: Can LMMs Comprehend Temporal and Contextual Narratives in Image Sequences?**
  Xiaochen Wang*, **Heming Xia***, Jialin Song, Longyu Guan, Yixin Yang, Qingxiu Dong, Weiyao Luo, Yifan Pu, Yiru Wang, Xiangdi Meng, Wenjie Li, Zhifang Sui
  Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing. ***EMNLP 2025 (Findings)****.*
- **SWIFT: On-the-Fly Self-Speculative Decoding for LLM Inference Acceleration**
  Heming Xia, Yongqi Li, Jun Zhang, Cunxiao Du, Wenjie Li
  The Thirteenth International Conference on Learning Representations. ***ICLR 2025****.*
- **Unlocking Efficiency in Large Language Model Inference: A Comprehensive Survey of Speculative Decoding**
  Heming Xia, Zhe Yang, Qingxiu Dong, Peiyi Wang, Yongqi Li, Tao Ge, Tianyu Liu, Wenjie Li, Zhifang Sui
  The 62nd Annual Meeting of the Association for Computational Linguistics. ***ACL 2024 (Findings)****.*
- **ImageNetVC: Zero- and Few-Shot Visual Commonsense Evaluation on 1000 ImageNet Categories**
  Heming Xia*, Qingxiu Dong*, Lei Li, Jingjing Xu, Tianyu Liu, Ziwei Qin, Zhifang Sui
  Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing. ***EMNLP 2023 (Findings)****.*
- **Bi-Drop: Enhancing Fine-tuning Generalization via Synchronous sub-net Estimation and Optimization**
  Shoujie Tong*, **Heming Xia***, Damai Dai, Runxin Xu, Tianyu Liu, Binghuai Lin, Yunbo Cao, Zhifang Sui
  Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing. ***EMNLP 2023 (Findings)****.*
- **Speculative Decoding: Exploiting Speculative Execution for Accelerating Seq2seq Generation**
  Heming Xia*, Tao Ge*, Peiyi Wang, Si-Qing Chen, Furu Wei, Zhifang Sui
  Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing. ***EMNLP 2023 (Findings)****.*
- **Enhancing Continual Relation Extraction via Classifier Decomposition**
  Heming Xia, Peiyi Wang, Tianyu Liu, Binghuai Lin, Yunbo Cao, Zhifang Sui
  The 61st Annual Meeting of the Association for Computational Linguistics. ***ACL 2023 (Findings, Short Paper)****.*

- **Improved deep learning techniques in gravitational-wave data analysis**
  Heming Xia, Lijing Shao, Junjie Zhao, Zhoujian Cao
  Physical Review D 103 (2021), 024040.

Other Publications

- **Reasoning Beyond Language: A Comprehensive Survey on Latent Chain-of-Thought Reasoning**
  Xinghao Chen*, Anhao Zhao*, **Heming Xia**, Xuan Lu, Hanlin Wang, Yanjun Chen, Wei Zhang, Jian Wang, Wenjie Li, Xiaoyu Shen
  Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing. ***EMNLP 2026 (Findings)***.
- **Finding RELIEF: Shaping Reasoning Behavior without Reasoning Supervision via Belief Engineering**
  Chak Tou Leong, Dingwei Chen, **Heming Xia**, Qingyu Yin, Sunbowen Lee, Jian Wang, Wenjie Li
  The 64th Annual Meeting of the Association for Computational Linguistics. ***ACL 2026 (Findings)***.
- **HauntAttack: When Attack Follows Reasoning as a Shadow**
  Jingyuan Ma*, Rui Li*, Zheng Li, Junfeng Liu, **Heming Xia**, Lei Sha, and Zhifang Sui
  The 64th Annual Meeting of the Association for Computational Linguistics. ***ACL 2026 (Findings)***.
- **From Query to Logic: Ontology-Driven Multi-Hop Reasoning in LLMs**
  Haonan Bian, Yutao Qi, Rui Yang, Yuanxi Che, Jiaqian Wang, **Heming Xia**, Ranran Zhen
  The 64th Annual Meeting of the Association for Computational Linguistics. ***ACL 2026 (Findings)***.
- **KNN-SSD: Enabling Dynamic Self-Speculative Decoding via Nearest Neighbor Layer Set Optimization**
  Mingbo Song, **Heming Xia**, Jun Zhang, Chak Tou Leong, Qiancheng Xu, Wenjie Li, Sujian Li
  The 19th Conference of the European Chapter of the Association for Computational Linguistics. ***EACL 2026 (Findings)***.
- **SpecVLM: Enhancing Speculative Decoding of Video LLMs via Verifier-Guided Token Pruning**
  Yicheng Ji*, Jun Zhang*, **Heming Xia**, Jinpeng Chen, Lidan Shou, Gang Chen, Huan Li
  Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing. ***EMNLP 2025***.
- **Towards Harmonized Uncertainty Estimation for Large Language Models**
  Rui Li, Jing Long, Muge Qi, **Heming Xia**, Lei Sha, Peiyi Wang, Zhifang Sui
  The 63rd Annual Meeting of the Association for Computational Linguistics. ***ACL 2025*** ***(Oral Presentation)***.
- **How Far are LLMs from Being Our Digital Twins? A Benchmark for Persona-Based Behavior Chain Simulation**
  Rui Li, **Heming Xia**, Xinfeng Yuan, Qingxiu Dong, Lei Sha, Wenjie Li, Zhifang Sui
  The 63rd Annual Meeting of the Association for Computational Linguistics. ***ACL 2025 (Findings)***.
- **PEToolLLM: Towards Personalized Tool Learning in Large Language Models**
  Qiancheng Xu, Yongqi Li, **Heming Xia**, Fan Liu, Min Yang, Wenjie Li
  The 63rd Annual Meeting of the Association for Computational Linguistics. ***ACL 2025 (Findings)***.
- **AppBench: Planning of Multiple APIs from Various APPs for Complex User Instruction**
  Hongru Wang, Rui Wang, Boyang Xue, **Heming Xia**, Jingtao Cao, Zeming Liu, Jeff Z. Pan, Kam-Fai Wong
  Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing. ***EMNLP 2024***.
- **A Survey on In-context Learning**
  Qingxiu Dong, Lei Li, Damai Dai, Ce Zheng, Jingyuan Ma, Rui Li, **Heming Xia**, Jingjing Xu, Zhiyong Wu, Baobao Chang, Xu Sun, Lei Li, Zhifang Sui
  Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing. ***EMNLP 2024***.
- **Enhancing Tool Retrieval with Iterative Feedback from Large Language Models**
  Qiancheng Xu, Yongqi Li, **Heming Xia**, Wenjie Li
  Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing. ***EMNLP 2024 (Findings)***.
- **Taking a Deep Breath: Enhancing Language Modeling of Large Language Models with Sentinel Tokens**
  Weiyao Luo, Suncong Zheng, **Heming Xia**, Weikang Wang, Yan Lei, Tianyu Liu, Shuang Chen, Zhifang Sui
  Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing. ***EMNLP 2024 (Findings)***.
- **Can Large Multimodal Models Uncover Deep Semantics Behind Images?**
  Yixin Yang, Zheng Li, Qingxiu Dong, **Heming Xia**, Zhifang Sui
  The 62nd Annual Meeting of the Association for Computational Linguistics. ***ACL 2024 (Findings)***.
- **Lossless Acceleration for Seq2seq Generation with Aggressive Decoding**
  Tao Ge, **Heming Xia***, Xin Sun*, Si-Qing Chen, Furu Wei
  Microsoft Research Technical Report.
- **Premise-based Multimodal Reasoning: Conditional Inference on Joint Textual and Visual Clues**
  Qingxiu Dong*, Ziwei Qin*, **Heming Xia**, Tian Feng, Shoujie Tong, Haoran Meng, Lin Xu, Zhongyu Wei, Weidong Zhan, Baobao Chang, Sujian Li, Tianyu Liu, Zhifang Sui
  The 60th Annual Meeting of the Association for Computational Linguistics. ***ACL 2022***.

Internship

JD.com (Tech Genius Team) **Apr. 2026 – Sep. 2026**
Research Intern *Beijing, China*
Mentor: Dr. Kelong Mao

Sea AI Lab **Jun. 2025 – Jan. 2026**
Research Intern *Singapore*
Mentor: Dr. Cunxiao Du

The Hong Kong Polytechnic University **Oct. 2023 – Jan. 2024**
Research Assistant at NLP Group *Hong Kong, China*
Advisor: Prof. Wenjie Li

Microsoft Research Asia **Oct. 2021 – Aug. 2022**
Research Intern at NLC Group *Beijing, China*
Mentor: Dr. Tao Ge

Services and Membership

- **Area Chair / Action Editor:** ICLR, ACL, EMNLP, COLM, ACL ARR
- **Reviewer / Program Committee Member:** ICML, ICLR, NeurIPS, ACL, EMNLP, NAACL, COLM, TMLR, ACM MM, WWW, EACL, AACL, TASLP, TMC, TWEB
- **Teaching Assistant:** COMP 5423: Natural Language Processing (Fall 2025 & Spring 2025), COMP 5140 (Fall 2024), COMP 2S01 (Spring 2024) at PolyU

Open-Source Projects

- **Reading List for Speculative Decoding (1.3k Stars★)**: Maintained a regularly updated paper list on Speculative Decoding, covering milestones, benchmarks, analytical studies, and applications on this promising research area.
- **Reading List for Efficient Reasoning (900 Stars★)**: Maintained a curated paper list on efficient reasoning, covering efficient training, latent/long-to-short CoT, adaptive thinking, optimal test-time scaling strategies, and more.
- **Spec-Bench for Speculative Decoding (Python, PyTorch, 400 Stars★)**: Developed a comprehensive benchmark and unified evaluation platform for assessing leading Speculative Decoding methods across diverse application scenarios.
- **Seq2Seq Inference Acceleration with Speculative Decoding (Python, Fairseq)**: Released all the code and checkpoints utilized in Speculative Decoding, which achieves 3x-5x inference speedup with only 300MiB of extra memory cost.

Invited Talks

- **NLP Group, King's College London**, TokenSkip: Controllable Chain-of-Thought Compression in LLMs, 11/2025.
- **NICE-NLP** and **MLNLP**, Sharing Panel - Efficient Reasoning in Large Language Models, 06/2025.
- **Huawei**, Hong Kong, Stop Overthinking: Towards Efficient Reasoning in Large Language Models, 05/2025.
- **COLING 2025 Tutorial**, Abu Dhabi, Speculative Decoding for Efficient LLM Inference, 01/2025.
- **CIP Group, CASIA**, Speculative Decoding: Past, Recent Advancements, and Future Directions, 04/2024.
- **NICE-NLP**, Unlocking Efficiency in LLM Inference: A Comprehensive Survey of Speculative Decoding, 03/2024.

Technical Skills

Languages: Python, LaTex, C/C++, Java, Shell, MATLAB, HTML/CSS
Developer Tools: VS Code, PyCharm, Git, Docker, Linux, Vim, Eclipse
Libraries/Frameworks: PyTorch, Transformers, Fairseq, TensorFlow, PyTorch-Lightning, spaCy, NumPy, WordPress

Honors and Awards

- Merit Student, Peking University 2021
- Scholarship of National Astronomical Observatory, Chinese Academy of Sciences 2019
- Merit Student, Henan Province, China 2016